

AYH MART: Checkout System

A Project On AYH MART - Checkout System Inclusive Integrated | Innovative

Programme
IMCA I Sem-2

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What we'll cover

Agenda

1. Project Abstract
2. Objective
3. System Description
4. Screenshots
5. Test Cases
6. References

Project abstract

AYH Mart checkout system

AYH Mart is a console-based checkout system developed in C, designed for a retail store. It allows customers to independently browse products, add items to a cart, and complete purchases via Cash or UPI QR Code payment.

The system supports three user roles - Customer, Staff, and Admin - each with a dedicated portal. Key features include product inventory management, a tiered membership discount program (Silver, Gold, Platinum), customer account management, and a detailed sales reporting module.

All data is managed in-memory using C structures, making the system lightweight, efficient, and fully functional as a retail management solution.



Project objectives



C-based system

Functional self-checkout system built in C for a real retail environment.



Role-based access

Separate access levels for Admin, Staff, and Customer users.



Inventory management

Add, Edit, Delete, and View products with cart and stock deduction.



Tiered membership

Silver (5%), Gold (10%), and Platinum (15%) discount tiers for members.



Dual payment

Cash with change calculation and UPI QR Code payment support.



Sales reports

Revenue, discount, and product-wise reports with structured data records.

System description

Languages, libraries, data structures and functions

Language & Platform

- C (Standard C)
- Windows Console App
- GCC / MinGW / Dev-C++
/ Code::Blocks

Libraries used

- stdio.h — I/O operations
- string.h — String handling
- ctype.h — Char classification
- stdlib.h — System calls, exit()
- windows.h — UTF-8 / Unicode support

Architecture

Menu-driven, modular function design.

5 core modules:

- Inventory
- Members
- Customer Self-Checkout
- Sales Reports
- Customer Accounts

Data structures (structs)

Product

pCode | pName | pPrice |
pStock | pCategory

Member

mId | mName | mSpent |
mTier | mDisc | mUsed

CartItem

code | name | price | qty

Sale & Customer

Sale: pCode | pName | qty | price
| total

Customer: cId | cName | cPass
| cMemberId

System description

Languages, libraries, data structures and functions

Functions

```
void mainManu();  
void admin();  
void adminDashboard();  
void inventory();
```

```
void add();  
void login();  
void view();  
void del();
```

```
void editProduct();  
void info();  
void cust();  
void custView();
```

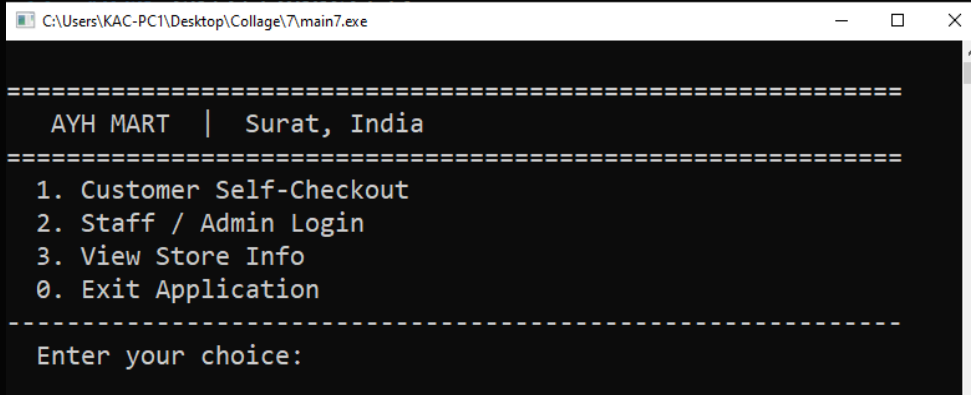
```
void members();  
void addMembers();  
void deleteMembers();  
void viewMembers();
```

```
void search();  
void addCart();  
void viewCart();  
void checkout();
```

```
void salesManu();  
void salesSummary();  
void productSales();  
void discountReport();
```

```
void recordSale(int, char[], int, float);  
float applyDiscount(int, float);  
void billing(float finalAmount, int isQR);  
void staff();  
void underDev();
```

Screenshots



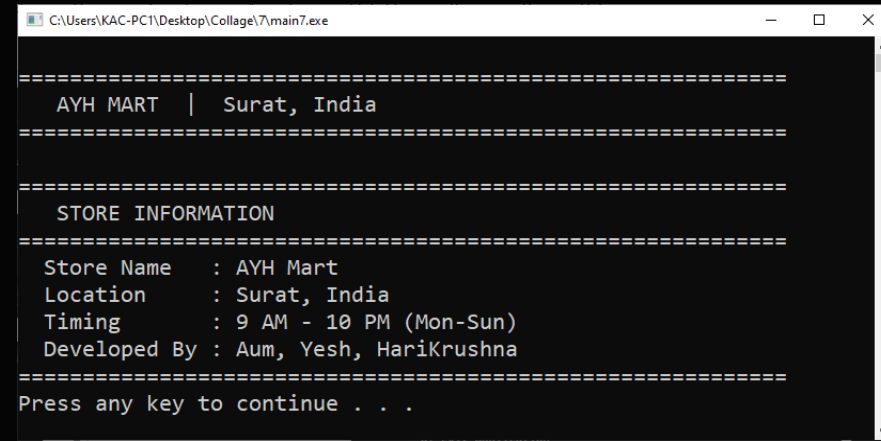
```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
  AYH MART |  Surat, India
=====
  1. Customer Self-Checkout
  2. Staff / Admin Login
  3. View Store Info
  0. Exit Application
=====
Enter your choice:
```

A screenshot of a Windows application window titled "C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe". The window displays a main menu for "AYH MART | Surat, India". The menu is enclosed in a box with dashed lines. The options listed are: 1. Customer Self-Checkout, 2. Staff / Admin Login, 3. View Store Info, and 0. Exit Application. Below the menu, it prompts the user to "Enter your choice:".

Main menu

View Store Info



```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
  AYH MART |  Surat, India
=====

=====
  STORE INFORMATION
=====
Store Name   : AYH Mart
Location    : Surat, India
Timing      : 9 AM - 10 PM (Mon-Sun)
Developed By : Aum, Yesh, HariKrushna
=====
Press any key to continue . . .
```

A screenshot of the same application window showing the "View Store Info" screen. The screen displays "STORE INFORMATION" in a box with dashed lines. The information shown is: Store Name: AYH Mart, Location: Surat, India, Timing: 9 AM - 10 PM (Mon-Sun), and Developed By: Aum, Yesh, HariKrushna. At the bottom, it prompts the user to "Press any key to continue . . .".

Screenshots



A screenshot of a terminal window titled "C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe". The terminal displays a menu for "AYH MART | Surat, India" separated by dashed lines. Below the menu, it shows a "LOGIN" section with the following text: "Username: admin", "Password: admin", "[] Logged in as ADMIN", and "Press any key to continue . . .".

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

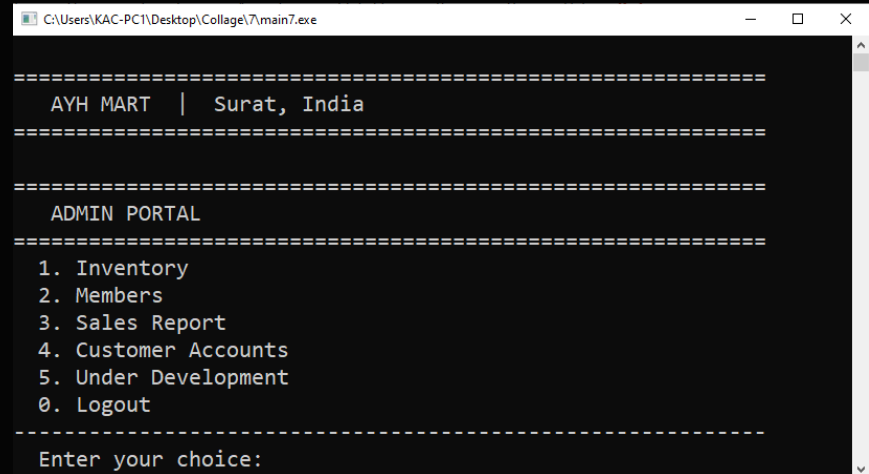
=====
AYH MART | Surat, India
=====

LOGIN
=====
Username: admin
Password: admin

[ ] Logged in as ADMIN
Press any key to continue . . .
```

Admin Portal

Admin Login



A screenshot of a terminal window titled "C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe". The terminal displays a menu for "AYH MART | Surat, India" separated by dashed lines. Below the menu, it shows an "ADMIN PORTAL" section with a list of options: "1. Inventory", "2. Members", "3. Sales Report", "4. Customer Accounts", "5. Under Development", and "0. Logout". At the bottom, it prompts "Enter your choice:".

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

ADMIN PORTAL
=====
1. Inventory
2. Members
3. Sales Report
4. Customer Accounts
5. Under Development
0. Logout
=====
Enter your choice:
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
INVENTORY MANAGEMENT
=====

1. Add Product(s)
2. Edit Product
3. Delete Product
4. View All Products
0. Back

=====
Enter your choice: _
```

Add Product(s)

Inventory Management

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

How many products to add: 2

-- Product 1 --
Code   : 106
Name    : aum
Price   : 222
Stock   : 3
Category: a

-- Product 2 --
Code    : 121
Name     : as
Price    : 21
Stock    : 2
Category: aas

[0] 2 product(s) added successfully.
Press any key to continue . . .
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
INVENTORY MANAGEMENT
=====

1. Add Product(s)
2. Edit Product
3. Delete Product
4. View All Products
0. Back

Enter your choice: _
```

Inventory Management

Edit Product

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

Enter Product Code to Edit: 121

Editing: as
Leave field blank / enter 0 to keep current value.

New Name [as]: aaa
New Price [21.00]: 20
New Stock [2]: 5
New Category [aas]: a

[0] Product updated successfully.
Press any key to continue . . . _
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
INVENTORY MANAGEMENT
=====

1. Add Product(s)
2. Edit Product
3. Delete Product
4. View All Products
0. Back

-----
Enter your choice: 3

Enter Product Code to Delete: 121
[0] Product deleted successfully.
Press any key to continue . . .
```

View all Product

Delete Product

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AVAILABLE PRODUCTS
=====

=====
CODE      NAME      PRICE     STOCK    CATEGORY
-----
101      Rice      45.00      80      Grocery
102      Milk      25.00     120      Dairy
103      Bread     30.00      60      Bakery
104      Shampoo   90.00      40      Personal
105      PenDrive  350.00     15      Electronics
106      aum       222.00      3       a

Press any key to continue . . .
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

MEMBER PORTAL
=====

1. Add New Member
2. Delete Member
3. View All Members
0. Back

-----
Enter your choice: █
```

Add New Member

Member Portal

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

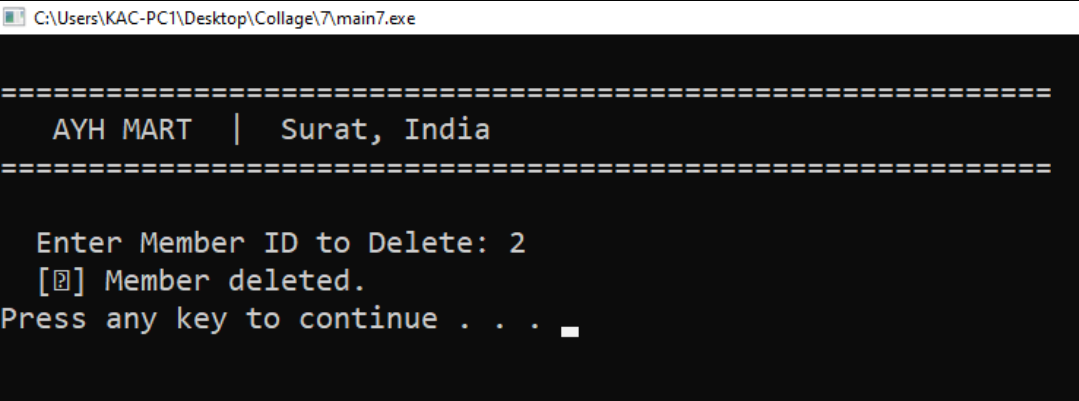
=====
AYH MART | Surat, India
=====

How many members to add: 1

-- Member 1 --
Member ID   : 1
Name        : aum
Amount Spent: 200
Tier (Silver/s | Gold/g | Platinum/p): s

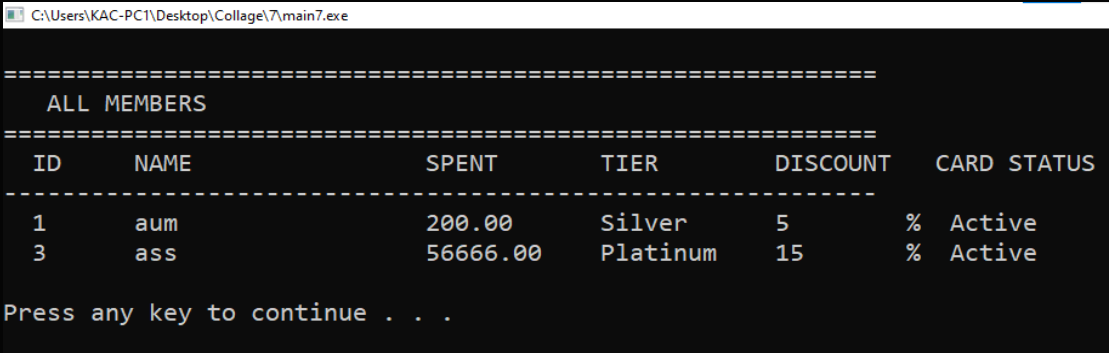
[0] Member(s) added successfully.
Press any key to continue . . . █
```

Screenshots



Delete Member

View all Member



Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

SALES REPORT
=====

1. Summary Report
2. Product-wise Sales
3. Member Discount Report
0. Back

Enter choice: █
```

Sale Summary

Sales Report

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

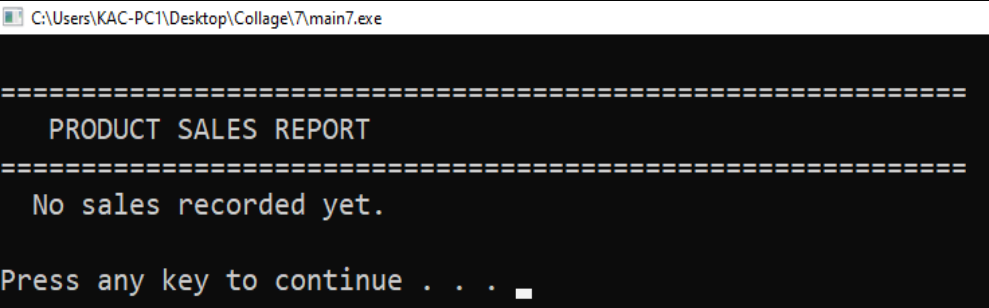
=====
AYH MART | Surat, India
=====

SALES SUMMARY
=====

Total Bills Generated : 0
-- Cash Payments      : 0
-- QR Payments        : 0
Total Items Sold      : 0
Total Revenue         : 0.00
Total Discount Given  : 0.00
Net Revenue           : 0.00
=====

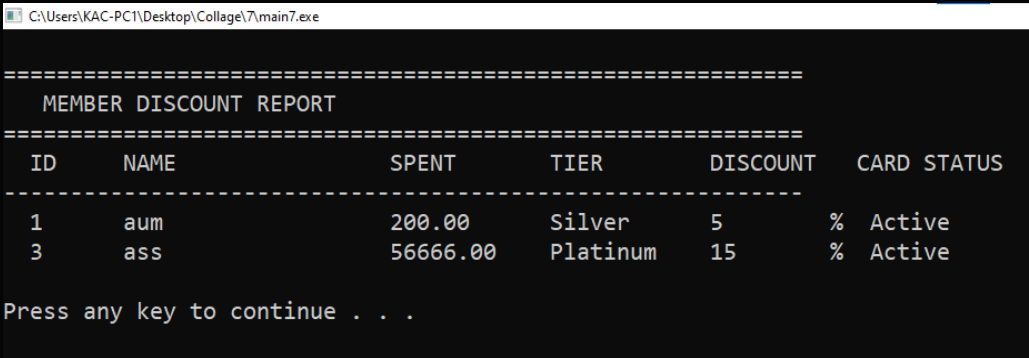
Press any key to continue . . . █
```

Screenshots



Product Sales Report

Member Discount Report



Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
CUSTOMER ACCOUNT MANAGEMENT
=====

1. Add New Customer Account
2. Delete Customer Account
3. View All Customer Accounts
0. Back

-----
Enter your choice: _
```

Customer Account Management

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

How many customer accounts to add: 3

-- Customer Account 1 --
Customer ID : 1
Name       : aum
Password   : 123
Confirm Pwd : 123
Member ID  : (enter 0 if no membership) 1

-- Customer Account 2 --
Customer ID : 2
Name       : as
Password   : 123
Confirm Pwd : 123
Member ID  : (enter 0 if no membership) 0

-- Customer Account 3 --
Customer ID : 3
Name       : ad
Password   : 123
Confirm Pwd : 123
Member ID  : (enter 0 if no membership) 3

[0] Customer account(s) created successfully.
Press any key to continue . . . _
```

Add New Customer Account

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
ALL CUSTOMER ACCOUNTS
=====
ID      NAME      PASSWORD  MEMBER ID
-----
1       aum       123       1
3       ad        123       3

Press any key to continue . . .
```

View all Customer

Delete Customer

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

Enter Customer ID to Delete: 2
[2] Customer account deleted.
Press any key to continue . . .
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
  AYH MART  |  Surat, India
=====

=====
  SYSTEM MODULES - UNDER DEVELOPMENT
=====

  1. Employee Management
  2. Stock Alerts / Reorder Levels
  3. Expense Tracking
  4. Backup & Data Security
  5. System Logs
  0. Back to Admin Portal
=====

Enter choice:
```

Under Development

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
CUSTOMER LOGIN
=====

Customer ID : 1
Password   : 123

Welcome, aum!
Press any key to continue . . .
```

Customer Portal

Customer Login

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
CUSTOMER SELF-CHECKOUT
=====

Welcome, aum

-----
1. View Store Products
2. Search for a Product
3. Add Item to Basket
4. View My Basket
5. Final Checkout
6. My Membership Card
0. Logout & Return to Main Menu
-----

Enter your choice:
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

=====
AVAILABLE PRODUCTS
=====
CODE      NAME      PRICE    STOCK   CATEGORY
-----
101      Rice      45.00     80     Grocery
102      Milk      25.00    120     Dairy
103      Bread     30.00     60     Bakery
104      Shampoo   90.00     40     Personal
105      PenDrive  350.00    15     Electronics
106      aum      222.00     3       a

Press any key to continue . . .
```

Search Product

View Product

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

Enter Product Code to Search: 1
[0] Product not found.

Press any key to continue . . .
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
  ADD ITEM TO BASKET
=====
CODE      NAME      PRICE    STOCK    CATEGORY
-----
101      Rice      45.00     80      Grocery
102      Milk      25.00    120      Dairy
103      Bread     30.00     60      Bakery
104      Shampoo   90.00     40      Personal
105      PenDrive  350.00    15      Electronics
106      aum      222.00     3        a

Enter Product Code : 101
Enter Quantity      : 5

+- Confirm Item -----+
| Name  : Rice          |
| Price : 45.00 per unit |
| Qty   : 5             |
| Total : 225.00        |
+-----+

Add to cart? (yes/no): y
[0] Item added to basket.

Press any key to continue . . . █
```

Add Item To Basket

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

MY BASKET
=====
CODE      NAME                PRICE    QTY    AMOUNT
-----
101      Rice                45.00     5    225.00
-----
Grand Total:                        225.00

Press any key to continue . . .
```

View Membership Card

View Basket

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

MY MEMBERSHIP CARD
=====
Account Holder : aum
Customer ID    : 1
-----
Member ID      : 1
Member Name    : aum
Tier           : Silver
Discount Rate  : 5%
Total Spent    : Rs. 200.00
Card Status    : Active (discount available)
-----

Press any key to continue . . .
```

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

Enter Member ID (0 if none): 1
[0] Membership discount applied (5%). Card is now marked as used.

-----
Subtotal      : 225.00
Discount      : 11.25
Payable Amount : 213.75
-----

SELECT PAYMENT METHOD:
1. Cash
2. QR Code (Digital)
Enter choice: 1
Pay by Cash (enter amount): 213.75

=====
[0] Payment received! Change: 0.00
Thank you for shopping at AYH Mart!
=====
Press any key to continue . . . █
```

Check-Out via cash

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
AYH MART | Surat, India
=====

Enter Member ID (0 if none): 0

-----
Subtotal      : 725.00
Discount      : 0.00
Payable Amount : 725.00
-----

SELECT PAYMENT METHOD:
1. Cash
2. QR Code (Digital)
Enter choice: 2

[ SCAN QR CODE TO PAY – Dhameliya Yash Mukeshbhai ]
```

```
+-----+
#####  ##  ##  ####  #####  #####
##      ##  #####  ##  #####  ##
##  #####  ##  ##  #####  #####  ##  #####  ##
##  #####  ##  ##  ##  #####  ##  #####  ##
##  #####  ##  ##  #####  #####  ##  #####  ##
#####  ##  ##  ##  ##  #####
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##  #####  ##  #####  ##  #####  #####  #####
##  #####  ##  #####  ##  #####  #####  #####
##  #####  ##  #####  ##  ##  ##  ##  #####
##  #####  ##  #####  #####  #####  ##  ##
#####  ##  #####  ##  #####  ##  ##
+-----+

UPI ID : 6353999107@ptyes
Amount : Rs. 725.00

=====
Transaction ID: AYH1000
Thank you for shopping at AYH Mart!
=====
Press any key to continue . . .
```

Check-Out via QR

Screenshots

```
C:\Users\KAC-PC1\Desktop\Collage\7\main7.exe

=====
  AYH MART |  Surat, India
=====
  1. Customer Self-Checkout
  2. Staff / Admin Login
  3. View Store Info
  0. Exit Application
-----
Enter your choice: 0

=====
  Thank you for visiting AYH Mart!
=====

-----
Process exited after 2945 seconds with return value 0
Press any key to continue . . .
```

Exit Application

Test cases

All 12 test cases passed. AYH MART Self Checkout System - functional validation results.

#	Test Case	Input	Expected Output	Result
01	Customer login - valid	Correct ID and password	"Welcome, Name!"	Pass
02	Customer login - invalid	Wrong ID or password	Invalid Customer ID or Password	Pass
03	Add to cart - valid product	Correct code, qty $\leq$ stock	Item added to basket	Pass
04	Add to cart - insufficient stock	qty > available stock	Insufficient stock warning	Pass
05	Membership discount - valid	Member ID with active card	Discount applied, card marked Used	Pass

References

Thank You

AYH MART
Checkout System